

The Retained-Distinction Tensor Field Theory: A Machine-Checked Discrete Spine PDE over the Rationals, with the Continuum Refused as a Non-Readout

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Abstract

We present the retained-distinction field — an amplitude over the nodes of a graph, tensor-valued once an internal index space is attached — as the solution of a single discrete master equation, a *spine* field PDE $M \partial_{tt} \Phi + D \partial_t \Phi - K L_R \Phi + \nabla V(\Phi) = J - \eta$, in which the spatial operator L_R is the root graph-Laplacian and every quantity is an exact rational readout. The theory is discrete on purpose. Its three named regimes — unitary oscillation, dissipative decay, and their union in the telegraph equation with mass $M = 1/(2\tau_c)$ — are machine-checked theorems, as is the discrete two-register dynamics that advances the field and the exact Euler-Lagrange envelope from which that dynamics is stationary. We prove that the telegraph discriminant $\text{disc} = D^2 - 4MK\lambda$ crosses zero at a single critical eigenvalue $\lambda_c = D^2/(4MK)$, and that this crossover is simultaneously the quantum/classical knife-edge (under-damped versus over-damped), the horizon of the audited core, and the agency threshold — one identity, three readings. We then give a complete, division-free discrete curvature chain over the rationals: Christoffel as first difference (affine yet flat), Riemann as second difference and as a Heisenberg commutator, both Bianchi identities, all four Riemann symmetries with pair symmetry *derived* not posited, a rational Levi-Civita connection needing only g^{-1} (no $\sqrt{g}$), and rational $SO(2)/SO(3)$ isometries. From the discreteness itself we then draw a single decisive falsifiable prediction: a bounded L_R -spectrum forces a positive *floor* on the eigenmode relaxation time τ_{rel} of bounded-spectrum modes (distinct from the mass constant $\tau_c = 1/(2M)$) that the unbounded continuum forbids, with a named datum that would retract it. Against the practice of discrete-gravity programmes, we do *not* pose a graph-to-continuum convergence gate. We replace it with a *Continuum Refusal*: the smooth field, the $\sqrt{g}$ orthonormal frame, and every $h \rightarrow 0$ limit are injected infinities that no agency reads, and are declined on thesis grounds — verified where refused, not left as gaps. The honest open remainder (the full $R^i{}_{jkl}$ in $n \geq 3$, the non-abelian group Bianchi in transport form, the general $SO(n)$ family, and absolute constant values — no anchor, none claimed) is enumerated by name and

kept strictly apart from what the continuum refusal declines. Every claim carries exactly one tier tag; the reader is invited to audit rather than believe.

Keywords: retained distinction; machine-checked proof (Coq); discrete tensor field theory; graph Laplacian; telegraph equation; quantum-classical crossover; horizon; discrete curvature; Bianchi identities; relaxation-time floor; falsifiable prediction; readout-not-truth; continuum refusal; reproducibility.

Preprint notice. This manuscript is a preprint and has not yet undergone formal peer review. Every machine-checked claim is nonetheless independently confirmable by recompiling the delivered Coq sources (Table 6, pinned by commit and hash prefix); interpretive readings and the continuum-refusal thesis remain provisional until independently reviewed.

1 Introduction and claim discipline

This paper makes one structural claim and defends it narrowly: a single discrete field equation over the rational numbers — the *spine* — carries, as exact and machine-checked readouts, the wave/quantum regime, the diffusive/classical regime, their telegraph union, a definite quantum-to-classical knife-edge that doubles as a horizon and an agency threshold, and a complete discrete curvature-tensor calculus with both Bianchi identities. The equation is tensor-valued and directly implementable as a finite-difference update; nothing in it takes a limit. That last point is not a limitation we apologise for. It is the thesis: **the discrete tensor PDE over $\mathbb{Q}$ is the theory**, and the continuum is a non-readout we decline (§7), never a target we chase.

The account has an unusual epistemic shape, which we state before any physics. The programme’s razor is *readout-not-truth*: an agency of finite means reads finite, rational, retained distinctions; the continuum — $\mathbb{R}$ -completeness, the $h \rightarrow 0$ limit, infinite scale separation, infinite precision — is an *injected infinity* that no reading ever returns. Wherever a construction would demand

Tag	Meaning
[Th.coqc]	Machine-checked over $\mathbb{Q}$; Print Assumptions = <i>Closed under the global context</i> ; coqc 8.20.1 exit 0; axiom-free.
[finite_diagnostic]	A finite diagnostic: an exact rational witness computed by Compute/vm_compute , not a universally-quantified proof.
[Dr]	Interpretive reading (physics gloss on a proven object); anchored on theorems but not itself a theorem.
[AX/import]	A disclosed identification or an imported result, cited at its own tier.
[Open]	A genuine positive extension, not yet proven — an honest research target, not a defect in a closed result.
[Refused]	A non-readout the theory <i>declines</i> on thesis grounds (readout-not-truth); <i>not</i> a gap. Verified where the refusal bites.

Table 1: The six claim tiers. Every claim in this paper carries exactly one. The load-bearing distinction is [Open] (unfinished, a target) versus [Refused] (declined, a position).

such an object we do one of two things: we exhibit the finite rational readout that does the work, or, where the infinite object is genuinely required and genuinely unread, we *refuse* it and prove where the refusal bites. Refusal is a position we verify, not a gap we hide.

Claim tiers. Every statement below carries exactly one of six tags, summarized in Table 1. The discipline is non-negotiable: a discrete result that is machine-checked is [Th.coqc] and is *never* written [Open]; the continuum and the $\sqrt{g}$ frame are [Refused] (a thesis position), never [Open] (a gap). We keep the [Open] remainder (§12) strictly apart from the [Refused] non-readouts, so that a reader can see at a glance which is unfinished work and which is declined on principle.

Gates, and the one we delete. A discrete field theory of gravity is usually asked to pass a ladder of gates. We keep the gates that ask whether the discrete object is *real* (exact, axiom-free, non-vacuous, witnessed) and we delete the gate that asks whether it *converges to a smooth field*. Table 2 lists our gates and their status; the deleted convergence gate appears in the last row, marked [Refused] and routed to §7. This is the single structural difference between the present programme and a continuum-seeking discrete-gravity programme, and it is deliberate.

What is not claimed. We do not claim a derivation of the continuum Einstein field equations, and we do not claim they are a limit our theory should reproduce; the gravitational content enters only through the discrete geometry operator $K L_R$ and the horizon knife-edge, both [Th.coqc]. We do not claim fixed numerical values for M, D, K, τ_c (these are [Open] — no anchor, none claimed; §12). We do not claim Born-rule uniqueness. Every non-theorem joint is disclosed and tagged.

Gate Asks	Status
G1 rational, no $\sqrt{}$ /division loss?	[Th.coqc]
Ex-act-ness	
G2 Print Assumptions Closed?	[Th.coqc]
Axiom-freedom	
G3 a nonzero-curvature witness exists?	[Th.coqc]
Non-vacuity	
G4 all four Riemann symmetries + both Bianchi?	[Th.coqc]
Sym-metry closure	
G5 a stationary discrete update (Euler–Lagrange)?	[Th.coqc]
Dynam-ics	
G6 one crossover = horizon = agency?	[Th.coqc]
Regime knife-edge	
G7 does it converge to a smooth field?	[Refused] (§7)
Graph-to-continuum	

Table 2: Gates for the discrete tensor theory. G1–G6 are the reality gates and all pass at [Th.coqc]. The classical G7 continuum-convergence gate is *deleted* and replaced by the Continuum Refusal (§7); it is not scored as pass or fail because it is not a criterion we accept.

2 The master spine as a tensor field PDE

2.1 The spine equation

Let G be a graph with node set $\mathcal{N}$ and let $\Phi : \mathcal{N} \rightarrow W$ be the retained-distinction field, valued in an internal index space W (tensor-valued once W carries indices). The field obeys the one master equation

$$M \partial_{tt} \Phi + D \partial_t \Phi - K L_R \Phi + \nabla V(\Phi) = J - \eta, \quad (1)$$

where M is inertia (the retained-distinction mass, $M = 1/(2\tau_c)$; §5), D is dissipation, K is the geometry coupling, L_R is the **root graph-Laplacian** — the retained-distinction operator, the discrete spatial-geometry axis — V is a self-interaction potential, J a source, and η a readout/back-reaction term. [[Dr] envelope over [Th.coqc] readouts.]

Setting $V \equiv 0$ and $J = \eta$ leaves the *spine*

$$M \partial_{tt} \Phi + D \partial_t \Phi + K L_R \Phi = 0, \quad (2)$$

whose three named readouts are theorems of the audited core:

Unitary oscillation. With $D \rightarrow 0$, (2) is the wave/quantum channel; the $M \partial_{tt}$ term drives oscillation (**InfoSchrodinger**). [Th.coqc]

Dissipative decay. With $M \rightarrow 0$, the $D \partial_t$ term drives diffusive decay (**InfoDecoherence**). [Th.coqc]

Telegraph union. The full (2) is the telegraph equation, with the exact rational mass identity $M = 1/(2\tau_c)$ (`InfoTelegraph`). [`Th.coqc`]

The spine identity across these readouts is machine-checked in the core; the tensor-PDE envelope (1) that reads them as one equation is the interpretive frame. [`Dr`]

2.2 Discrete dynamics — the actual update

Equation (1) is not a shorthand for a continuum operator. On an L_R -eigenmode of eigenvalue λ and a lattice tick, ∂_{tt} and ∂_t are finite differences and the field is advanced by an exact two-register (position/velocity) update:

$$X[n+1] = X[n] + \Delta\theta_R V[n], \quad (3)$$

$$V[n+1] = V[n] + \frac{\Delta\theta_R}{M} \left(K L_R \Phi - D V[n] - \nabla V(\Phi) + J - \text{carr} \right) \quad (4)$$

Here $n \in \mathbb{N}$ is the tick counter and $\Delta\theta_R$ is the retained-distinction phase increment — *not* a physical dt , and never the limit $dt \rightarrow 0$ (see the Clock Registry, §3). Every register is a rational readout; no limit is taken. The backward-Euler / decay-ratio form of the $D \partial_t$ piece is the core brick `InfoDecayRatioFromTauC`; its continuum companion $\exp(-\Gamma_{\text{damp}} t)$ is an $h \rightarrow 0$ object (I2) and is [`Refused`] (§7). [`Th.coqc`]

2.3 Action and Euler–Lagrange envelope

The update (3)–(4) is not posited; it is stationary for a discrete window action. Over one node-window $p \rightarrow m \rightarrow q$,

$$A_{\text{win}}(m) = M |q-m|^2 + M |m-p|^2 - h^2 K \text{gform}(m, m), \quad (5)$$

with linear coefficient $S_{\text{geo}} = h^2 K \text{gform}(m, m) - \sum_e \text{benefit}(e)$ and diagonal $C_i = 2M - h^2 K \text{gdeg}(i)$. Stationarity $\partial A_{\text{win}} / \partial m = 0$ reproduces the discrete spine update exactly, as a ring identity in $\mathbb{Q}$ — no variation-in-the-limit is invoked (`InfoActionStationarity`). The geodesic companion gives $M \text{acc} = \frac{1}{2} \gamma \text{vel}^2$ as the exact discretization of $M \ddot{x} = -\frac{1}{2} M' \dot{x}^2$ (`InfoGeodesicActionStationarity`). [`Th.coqc`]

3 Clock Registry

Time-like symbols are the classic site of silent substitution: a tick becomes a duration, a phase increment becomes a metric interval, a limit is slipped in. We firewall every time-like symbol in Table 3. The rule is one line: *no symbol below is silently substituted for another, and each equation states which clock it runs on.* [`Dr`]

4 Node Registry

Table 4 lists the chain as a registry of nodes: each an equation or object, with its file/theorem, its tier, and

the gate it must pass. The flow is *difference* $\rightarrow L_R \rightarrow \text{spine} \rightarrow \text{telegraph} \rightarrow \text{crossover} \rightarrow \text{horizon} \rightarrow \text{curvature} \rightarrow \text{Bianchi}$. Every node but the last-listed frame refusal is [`Th.coqc`]; N14 is a [`Th.coqc`] theorem *that* the $\sqrt{g}$ frame is a non-readout, i.e. a verified refusal.

5 The telegraph discriminant crossover: quantum $\leftrightarrow$ classical = horizon = agency knife-edge

The temporal characteristic of the spine (2) on an L_R -eigenmode λ is

$$\text{pchar}(\omega) = M\omega^2 + D\omega + K\lambda, \quad \text{disc} = D^2 - 4MK\lambda, \quad (6)$$

with crossover $\lambda_c = D^2/(4MK)$. The entire crossover is carried by one algebraic identity, the completion of the square:

$$4M \cdot \text{pchar}(\omega) = (2M\omega + D)^2 - \text{disc} \quad (\text{char_complete_square}, [\text{Th.coqc}]) \quad (7)$$

From (7) the two regimes and the edge follow exactly over $\mathbb{Q}$:

- **Under-damped = quantum** ($\text{disc} < 0$): $\text{pchar}(\omega) > 0$ for every real ω , so there is no real frequency root and the mode is oscillatory — the wave/quantum (`InfoSchrodinger`) side (`underdamped_positive`). [`Th.coqc`]
- **Over-damped = classical** ($\text{disc} \geq 0$): $\text{pchar}(\omega) \leq 0$ at $\omega = -D/2M$, so a real decay root exists and the mode purely decays — the diffusive/classical (`InfoDecoherence`) side (`overdamped_min_nonpositive`). [`Th.coqc`]
- **Crossover / critical damping**: $\text{disc}(\lambda_c) = 0$ at $\lambda_c = D^2/(4MK)$ (`critical_disc_zero`); and $\text{disc} < 0 \iff D^2 < 4MK\lambda$, i.e. λ above λ_c (`underdamped_iff_above_crit`). [`Th.coqc`]
- **Witnesses**: $(M, D, K, \lambda) = (1, 1, 1, 1) \Rightarrow \text{disc} = -3 < 0$ (under-damped/quantum); $(1, 4, 1, 1) \Rightarrow \text{disc} = 12 > 0$ (over-damped/classical). [`finite_diagnostic`]

One knife-edge, three readings. The telegraph discriminant (6) is the audited core's `InfoFrontier.spine_discr`, and λ_c is `spine_lambda_c`; the oscillatory (quantum) regime is exactly the region above the horizon (`disc_is_spine_discr`, `lam_c_is_spine_lambda_c`, `quantum_iff_above_horizon`, `oscillatory_above_horizon`, `InfoTelegraphHorizonUnification`). [`Th.coqc`]. The reading: a single rational knife-edge λ_c unifies the quantum $\leftrightarrow$ classical transition (dynamics), the black-hole horizon (the geometric side of the core), and the agency threshold — looping/quantum above it, settling/classical at or below it. [`Dr`]

Symbol	Name	What it is	Must not be confused with
n	tick index	dimensionless counter of update steps (N)	physical elapsed time t ; a duration
$\Delta\theta_R$	retained-distinction phase increment	the per-tick advance in (3)–(4)	a physical dt ; a metric interval
τ_c	correlation / agency time	the retained-distinction correlation scale; the mass constant, $M = 1/(2\tau_c)$	τ_{rel} (eigenmode relaxation time); τ_R ; the horizon λ_c ; wall-clock t
dt	observer step	a chosen finite readout step (backward-Euler in <code>InfoDecayRatioFromTauC</code>)	$\Delta\theta_R$; the continuum limit $dt \rightarrow 0$ ([Refused])
t	physical time	the interpretive elapsed-time readout $t = n \cdot dt$	the tick index n ; $\Delta\theta_R$
Γ_{damp}	damping rate	the dissipative rate $\Gamma_{\text{damp}} = D/(2M)$ (over-damped exponent)	Γ_{rel} ; τ_c ; λ_c ; a frequency ω
$\Gamma_{\text{rel}}(\lambda)$	eigenmode relaxation rate	the per-mode relaxation rate $\Gamma_{\text{rel}} = K\lambda/D$ (§8)	$\Gamma_{\text{damp}} = D/(2M)$; the mass scale τ_c
$\tau_{\text{rel}}(\lambda)$	eigenmode relaxation time	the per-mode relaxation time $\tau_{\text{rel}} = 1/\Gamma_{\text{rel}} = D/(K\lambda)$; floored by a bounded spectrum	τ_c (the mass scale, $M = 1/(2\tau_c)$); τ_R ; the horizon λ_c
τ_R	readout / refresh time	the observer's readout cadence	τ_c (a system property, not the observer's)
ω	mode frequency	the temporal characteristic root of $M\omega^2 + D\omega + K\lambda = 0$	Γ_{damp} (decay) — the sign of the discriminant decides which

Table 3: Clock Registry. Every time-like symbol in the paper, with what it is and the symbols it must never be silently swapped for. The continuum limit $dt \rightarrow 0$ appears only as an explicitly [Refused] object.

6 The discrete curvature-tensor chain

The geometry operator L_R in (1) is the entry point to a full discrete curvature calculus, division-free and exact over $\mathbb{Q}$. Every step is [Th-coqc].

1. Christoffel as first difference. `christoffel_fd w j = w(Sj) - w(j)`. An affine weight gives a nonzero Christoffel (= b) yet zero curvature: Christoffel $\neq$ curvature, exactly (`christoffel.can.be.nonzero.while.flat`).

2. Riemann as second difference. `riemann_fd = w(SSj) - 2w(Sj) + w(j)`; affine $\Rightarrow$ flat; the non-affine witness $w = qn^2$ gives `riemann_fd 0 = 2  $\neq$  0`; and flat $\iff$ locally affine (arithmetic progression) (`riemann.is.second.difference, affine.is.flat, curvature.nonzero.witness, flat.iff.second.diff.zero`).

3. Riemann as commutator (genuinely 2-index). On the Heisenberg / upper-unipotent group with polynomial (division-free) inverse, R_{xy} is the center of $[X, Y] = a \cdot b$, with $[Y, X] = -a \cdot b$ (kl -antisymmetry, exact in the nilpotent center); abelian, scalar, and same-direction cases are all flat, strictly exceeding the scalar brick (`plaquette.curvature.z, reverse.antisymmetric, curvature.is.central`).

4. First (algebraic) Bianchi and ij -antisymmetry. Metric-compatible $\mathfrak{so}(3) \cong (\mathbb{Q}^3, \times)$: the generator `asym u` is antisymmetric, the matrix commutator is `asym(u  $\times$  w)` (so $\mathfrak{so}(3)$ is closed under bracket), curvature is ij -antisymmetric, and the Jacobi identity is the first Bianchi identity (`asym.is.antisymmetric, commutator.is.asym.cross, curvature.ij.antisymmetric, bianchi.jacobi`).

5. Second (differential) Bianchi. On the 3-D lattice, abelian $dF = 0$: $d_z F_{xy} + d_x F_{yz} + d_y F_{zx} = 0$, from mixed-difference commutation, with a nonzero-curvature witness so the identity is non-vacuous (`second.bianchi, mixed.differences commute`).

6. Pair symmetry, derived. $R_{ijkl} = R_{klij}$ is derived — not posited — from A1 (ij -antisymmetry) + A2 (kl -antisymmetry) + B1 (first Bianchi), for abstract R over $\mathbb{Q}$, via the classical four-Bianchi-sum argument (`pair.symmetry`). All four Riemann symmetries are thus machine-checked.

7. Metric-derived Levi-Civita (rational). $\Gamma^2_{12} = \frac{1}{2}dw/w$ is rational for $w \neq 0$ (needing only g^{-1} , no $\sqrt{g}$, no I1); $R_{1212} = -\frac{1}{2}d^2w + (dw)^2/(4w)$; flat on constant w , curved witness $R_{1212} = -\frac{1}{4}$ (`christoffel.G212.rational, flat.constant.zero.curvature, curved.witness`).

8. Rational isometry group. $SO(2)$ exists over $\mathbb{Q}$ via the Cayley/Pythagorean transform $N^T N = (1 + t^2)^2 I$; a rational $SO(3)$ capstone (two (3, 4, 5) rotations, holonomy $\neq I$) gives nonzero rational curvature, division-free (`InfoRationalIsometry, InfoRationalS03Curvature`).

All four Riemann symmetries (kl -antisymmetry, ij -antisymmetry, first Bianchi, pair symmetry) and both Bianchi identities are machine-checked; the discrete curvature chain is *structurally complete* over $\mathbb{Q}$. [Th-coqc]

7 Continuum Refusal

A continuum-seeking discrete-gravity programme poses a graph-to-continuum gate (G7, Table 2): does the discrete theory converge to a smooth tensor field? **We delete that gate and replace it with a Continuum Refusal.** The discrete tensor PDE over $\mathbb{Q}$ (§2) is the

theory; there is no smooth limit to reach, and reaching one is not a success criterion. [Refused] (thesis).

The diagnosis is readout-not-truth: every continuum object the deleted gate would demand is an *injected infinity*, and therefore a non-readout that an agency of finite means never returns.

- **I1 ($\mathbb{R}$ -completeness)** — the smooth field; the orthonormal frame $\sqrt{g}$. Proven refused: the rational metric $\text{diag}(1,2)$ has *no* rational frame, so $\sqrt{g}$ is I1, while the connection and curvature stay rational readouts (`InfoMetricFrameNonReadout` · `metric_frame_diag12_no_rational`). [Th_coqc]→[Refused]
- **I2 ($h \rightarrow 0$ / Taylor limit)** — continuum d^2g , $d\Gamma$, the exponential $\exp(-\Gamma_{\text{damp}}t)$. Our second difference and backward-Euler are the finite readouts; the limit is declined (`InfoDecayRatioFromTauC`). [Refused]
- **I3/I4 ($\text{Re} \rightarrow \infty$, $+\infty$ tails)** — not invoked; refused where they would appear. [Refused]

The Continuum Refusal is a *position*, verified where it bites, not an unmet goal. We never say the theory “converges to” or “should reproduce” a continuum tensor PDE. The discrete PDE is complete on its own terms; the continuum is the map, not the territory. Crucially, refusal is not a synonym for open. An I1 object like $\sqrt{g}$ is not a theorem we have failed to prove — we have proven it is not a rational readout, and we decline it on principle. This is the single most important distinction in the paper, and §12 keeps it strictly apart from the genuine open work.

8 The decisive falsifiable prediction: a τ_{rel} floor from a bounded spectrum

This is the tier-jump. Everything up to here is an *internal* theorem about a discrete kernel; here the discrete kernel forces a statement about the *world* that the continuum cannot make, that has the opposite sign, and that a single clean measurement decides. We do not hide behind “interpretation”: we write the inequality, we name the arenas, and we pre-commit to the retraction that a violating datum compels.

8.1 The structure: discreteness $\Rightarrow$ a rate ceiling $\Leftrightarrow$ a relaxation-time floor

Our framework lives on a *discrete* graph — atomicity / retained distinction is the root, so the readout operator L_R acts on a finite (or locally finite) index set. A finite symmetric L_R has a *bounded* spectrum: every eigenvalue obeys

$$0 \leq \lambda \leq \Lambda, \quad \Lambda \leq 2 \max_i \deg(i) \quad (8)$$

by Gershgorin — the largest L_R -eigenvalue is capped by twice the maximum node degree. Atomicity is not a

decorative postulate here: it is exactly the hypothesis $\lambda \leq \Lambda$.

On an L_R -eigenmode the spine (2) gives an *eigenmode relaxation rate* $\Gamma_{\text{rel}}(\lambda)$ obeying $\Gamma_{\text{rel}} D = K\lambda$, and a corresponding *eigenmode relaxation time* $\tau_{\text{rel}}(\lambda) = 1/\Gamma_{\text{rel}} = D/(K\lambda)$. This relaxation time is a property of the bounded-spectrum mode; it is *not* the correlation/agency time $\tau_c = 1/(2M)$ of the Clock Registry (§3), nor the damping rate $\Gamma_{\text{damp}} = D/(2M)$, and the three must not be conflated. Feeding the bounded spectrum (8) into $\Gamma_{\text{rel}} D = K\lambda$ yields, exactly over $\mathbb{Q}$, a *ceiling* on the relaxation rate and therefore a *positive floor* on the eigenmode relaxation time:

$$\Gamma_{\text{rel}} \leq \frac{K\Lambda}{D} \quad \Leftrightarrow \quad \tau_{\text{rel}} = \frac{D}{K\lambda} \geq \frac{D}{K\Lambda} > 0 \quad (9)$$

The ceiling half is machine-checked with no axioms:

Theorem 1 (Discrete rate ceiling / τ_{rel} floor, [Th_coqc]). *For rationals $\Gamma_{\text{rel}}, \lambda, \Lambda, K, D$ with $\Gamma_{\text{rel}} D = K\lambda$, $\lambda \leq \Lambda$ and $K \geq 0$, one has $\Gamma_{\text{rel}} D \leq K\Lambda$ (`InfoTauFloor.discrete_rate_ceiling`). Consequently a bounded discrete spectrum bounds the energy as well, $E^2 M \leq \hbar^2 K\Lambda$ (`InfoHypothesisSeeds.energy_uv_bounded`): the same atomicity that floors τ_{rel} is the built-in UV cutoff.*

The floor is not a soft “typical scale”; it is a hard lower bound that holds *no matter how hard the system is driven*, because Λ is fixed by the graph, not by the drive.

8.2 The continuum has no floor — the two are mutually exclusive

The continuum quantum speed limit (Mandelstam–Tamm / Margolus–Levitin) reads $\tau_{\text{rel}} = \pi\hbar/(2E)$ (equivalently $\tau_{\text{rel}} = \hbar/(2\Delta E)$). Its spectrum is *unbounded*: there is no largest eigenvalue, so raising E (or ΔE) without limit drives $\tau_{\text{rel}} \rightarrow 0$. There is *no* positive floor. This too is machine-checked:

Theorem 2 (Continuum: no floor, [Th_coqc]). *For any rational bound B and any $K \neq 0$ there exists an eigenvalue λ with $K\lambda > B$ (`InfoTauFloor.continuum_no_floor`): the rate $K\lambda$ exceeds every bound, i.e. τ_{rel} can be made arbitrarily small.*

Theorems 1 and 2 are the same statement with opposite quantifiers on the spectrum: *bounded* $\Rightarrow$ floor, *unbounded* $\Rightarrow$ no floor. They are logically *mutually exclusive* about the world. A physical system either exhibits a τ_{rel} that *saturates* at a positive value as E (or the coupling) is pushed up — discreteness — or a τ_{rel} that keeps shrinking $\propto 1/E$ with no floor — unbounded continuum. One clean curve of τ_{rel} versus E separates them. We do not need to “believe” either side; we audit the curve.

8.3 Real arenas and order-of-magnitude estimates

The floor magnitude $\tau_{\text{rel}}^{\min} = D/(K\Lambda)$ is *system-dependent*: it is set by that system’s independently

measurable spectral ceiling Λ (a bandwidth, a gap, a maximum coupling), converted to an energy $E_\Lambda = \hbar\sqrt{K\Lambda/M}$. The *structure* (9) is [Th.coqc]; the *numeric magnitude* of any one arena’s floor is [finite_diagnostic]/[Dr] — a measured or estimated rational, stated as such, never promoted to a theorem. Order-of-magnitude candidates, each with a ceiling that can be measured on the *same* sample by an independent method:

- **Circuit / cavity QED — maximum Rabi rate and gap.** The vacuum Rabi splitting $2g$, the transmon gap, and the anharmonicity all cap the achievable transition rate. With gaps of order a few $\times 10^9$ – 10^{10} Hz, the floor sits near $\tau_{\text{rel}}^{\text{min}} \sim 10^{-10}$ – 10^{-11} s. Ultra-strong-coupling devices are the place to push the drive until the rate *stops* rising: continuum QSL says it keeps rising, the floor says it saturates at the gap-set value. [Dr]/[finite_diagnostic]
- **Solid-state phonon readout — the Debye ceiling.** A lattice has a maximum phonon frequency $\omega_D \sim 10^{13}$ – 10^{14} rad s $^{-1}$; that is a literal Λ (the top of the acoustic band). Phonon-mediated decoherence/relaxation cannot run faster than that band, giving $\tau_{\text{rel}}^{\text{min}} \sim 10^{-13}$ – 10^{-14} s independent of drive strength. ω_D is measured independently (neutron/X-ray phonon dispersion, heat capacity), which is exactly the “independently-measured Λ ” the test requires. [Dr]/[finite_diagnostic]
- **Electronic bandwidth — the attosecond frontier.** An electronic band of width $W \sim 1$ – 10 eV sets $\tau_{\text{rel}}^{\text{min}} \sim \hbar/W \sim 10^{-16}$ – 10^{-17} s. Attosecond streaking and pump–probe metrology already operate at tens of attoseconds ($\sim 10^{-17}$ s) — right at this floor. The decisive extension is to push band-limited electronic dynamics toward the zeptosecond regime (10^{-21} s): the continuum QSL permits it at high enough photon energy, the discrete floor forbids it for a fixed bandwidth. [Dr]/[finite_diagnostic]
- **Margolus–Levitin saturation.** Systems engineered to saturate the ML bound $\tau_\perp = \pi\hbar/2E$ (maximal parallelism) are the sharpest probe: continuum ML keeps $\tau_\perp$ falling as $1/E$ without limit, whereas the discrete framework predicts the saturated ML curve itself *bends over* to a plateau once E approaches E_Λ . A measured ML curve that plateaus confirms a ceiling; one that keeps falling straight through E_Λ falsifies us. [Dr]

The common experimental signature across all four: hold the system fixed, measure its spectral ceiling Λ (bandwidth / gap / max Rabi / ω_D) by an *independent* method, then drive the transition/decoherence ever harder and plot $\tau_{\text{rel}}(E)$. A plateau at $\tau_{\text{rel}}^{\text{min}} = D/(K\Lambda)$ is the discrete signature; a straight $1/E$ descent with no plateau is the continuum signature. The four arenas and their independently-measured ceilings are collected in Table 5.

8.4 The confirming datum, the falsifying datum, and the retraction commitment

Confirmation. A system whose τ_{rel} *saturates* at a positive floor, with the floor value tracking an *independently measured* spectral ceiling Λ across several samples (change Λ , watch $\tau_{\text{rel}}^{\text{min}}$ move as $D/(K\Lambda)$), confirms the discrete prediction (9). This is genuine confirmation, not curve-fitting, precisely because Λ is fixed by a *separate* measurement, so the floor is predicted, not tuned.

Falsification. A single, reproducible measurement of τ_{rel} (equivalently, a transition/orthogonalization/decoherence rate Γ_{rel}) that lies *below* the floor $D/(K\Lambda)$ — i.e. $\Gamma_{\text{rel}} > K\Lambda/D$ — for a system whose Λ has been independently established, *falsifies the discrete framework*. There is no free parameter to absorb it: D , K are the same constants that fix every other spine readout, and Λ is measured, not fitted. A verified sub-floor rate says the world’s spectrum is unbounded where we claimed it was bounded — the continuum wins.

Retraction commitment (explicit). If such a below-floor datum is independently reproduced with a defensibly measured Λ , we will: (i) mark Theorem 1’s *physical application* [Refused] (the theorem itself stays true of the kernel; what fails is the world-identification); (ii) withdraw the atomicity-as-UV-cutoff reading (InfoHypothesisSeeds.energy_uv_bounded) as a claim about nature; (iii) record the falsifying reference, the measured Λ , and the observed $\Gamma_{\text{rel}} > K\Lambda/D$ in the manuscript erratum and the repository RETRACTIONS log, pinned by commit; and (iv) demote the whole discrete-floor programme from a prediction to a refuted hypothesis in the claim ledger. We state this *before* the measurement so that the prediction is falsifiable in the strict sense: we have named, in advance, the observation that would end it.

This is the honest tier-jump: not that the discrete kernel is *true*, but that it says something [Th.coqc]-grounded, quantitative, and *different* from the continuum — a τ_{rel} floor where continuum QM/GR guarantees none — and we have written down the number, the arena, and the datum that would refute it.

9 Positioning against related programmes

This programme sits at the confluence of three mature literatures — discrete and emergent gravity, machine-checked mathematics and physics, and the informational reconstruction of quantum mechanics — and borrows the ambition of each without claiming their depth. We state up front the honest asymmetry: *on its own turf, every programme cited below goes further than we do*. Our single structural contribution is not to out-reach any of them but to hold, in one machine-checked kernel

over $\mathbb{Q}$, the join they individually leave open (§10). We map the terrain first.

Discrete and emergent gravity. A family of programmes takes spacetime, or its geometry, to be built from more primitive discrete or relational data. Causal-set theory grows a Lorentzian manifold as the continuum approximation of a locally finite partial order; Regge calculus and its dynamical-triangulation descendants (Euclidean DT and Causal Dynamical Triangulations) discretize the Einstein–Hilbert action on simplicial complexes and recover smooth geometry in a scaling limit; loop quantum gravity quantizes geometry into discrete spin-network area and volume spectra; quantum graphity and related condensed-matter-of-spacetime models let a background graph dynamically order into a geometric phase; deterministic cellular-automaton and hypergraph-rewriting programmes (the cellular-automaton interpretation of quantum mechanics; the hypergraph rewriting approach) reconstruct both quantum amplitudes and gravitational curvature from local update rules on a discrete substrate. We share these programmes’ starting intuition — geometry and dynamics as readouts of a discrete relational structure — and we share, in particular, the hypergraph/automaton instinct that *one local update rule* should carry both quantum behaviour and curvature. *They go deeper than we do* on their own ground: each has a far richer kinematics, a continuum-limit programme, and in several cases phenomenological reach that we do not attempt. The one place we diverge sharply is the continuum: where these programmes seek, gate on, or conjecture a smooth limit, we delete that gate and refuse the limit as a non-readout (§7). [Dr]

Machine-checked mathematics and physics. A second literature supplies our method rather than our subject. Large formalization efforts — the machine-checked four-colour and Feit–Thompson (odd-order) theorems, the formal proof of the Kepler conjecture, and the formalization of relativistic kinematics and Minkowski spacetime in Isabelle/HOL, alongside verified numerical-analysis and PDE developments — have established that deep mathematical and physical content can be carried end-to-end by a proof kernel with no trusted human step. We adopt exactly this standard, and adopt its discipline literally: axiom-free over $\mathbb{Q}$, `Print Assumptions Closed under the global context`, `coqc exit 0` (§13). *These efforts go far deeper than ours* in scale and in the sophistication of what they mechanize; our formal corpus is small by their measure. What we take from them is not size but the non-negotiable bar — and we apply that bar to a physics-shaped object (a discrete field PDE and its curvature calculus) rather than to a pure-mathematics theorem. [Dr]

Emergent and thermodynamic gravity. A third literature reads the gravitational field equations as an equation of state rather than a fundamental law: the Bekenstein–Hawking horizon entropy; the derivation of the Einstein equations from the Clausius relation across

local horizons; the entropic-force account of gravity; and the programme reading gravitational dynamics off horizon thermodynamics and holographic degrees of freedom. Our horizon knife-edge λ_c (§5) is thesis-adjacent to this literature: it makes a single rational threshold do the work of a horizon, a quantum/classical divide, and an agency boundary, in the spirit of “geometry is a coarse readout of something informational.” *This literature goes much deeper* — it recovers the full continuum field equations and quantitative thermodynamic relations, which we neither derive nor target. Our knife-edge is a machine-checked discrete identity, not a thermodynamic derivation of dynamical gravity, and we mark that gap plainly (§10, §11). [Dr]

Informational reconstructions of quantum theory. A fourth literature derives the quantum formalism from operational or informational axioms: the reconstruction of quantum theory from five reasonable axioms; the derivation of finite-dimensional quantum theory from informational principles (purification and companions); the characterization of quantum mechanics through no-signalling and no-broadcasting in a C^* -algebraic setting; and the relational reading in which quantum states encode an observer’s information about a system. Our spine shares the conviction that the quantum branch is a *readout* of an information-bearing structure rather than a fundamental ontology, and our under-damped/oscillatory regime plays the role their reconstructed quantum kinematics plays. *These reconstructions go deeper than we do* on the quantum side: they recover the full Hilbert-space formalism, the tensor-product rule, and (in some) the Born rule, none of which we derive — indeed we explicitly refuse Born-rule uniqueness as a non-readout (§12). What we add is not a better reconstruction but a shared kernel in which the quantum readout and the curvature readout are branches of *one* machine-checked equation. [Dr]

10 A falsifiable absence claim

Having granted that each neighbouring programme is deeper on its own turf, we can state precisely the one thing we believe is not yet present in any of them, and we state it as a *falsifiable absence claim* in the sense of a search we invite the reader to defeat.

Claim C1 (falsifiable absence). *We know of no single machine-checked equation — axiom-free, recom-pilable to a proof-kernel exit 0 — whose readouts are both the quantum-dynamics branch and the discrete-curvature/gravity branch.* More sharply: lattice quantum field theory computes the quantum plus special-relativistic content on a fixed discrete background, but its lattice carries no dynamical curvature; and the large unification programmes that *do* join quantum behaviour to dynamical geometry (§9) have, to our knowledge, no verified kernel — no single mechanically-checked object from which both branches issue as theorems. The join “quantum readout *and* curvature readout, from one axiom-free equation, machine-checked” is the cell

we claim is empty. [Open] (an absence, held open to refutation).

Why this is a claim and not a boast. C1 asserts an *absence*, and an absence is the most falsifiable kind of statement: a single counterexample retires it. We therefore attach the search that backs it and the commitment that governs it, so that the claim is testable rather than rhetorical.

Search protocol (stated so it can be repeated). We looked for a counterexample along three axes and report the boundary in each.

- **Discrete-gravity programmes** (causal sets, Regge/CDT, loop quantum gravity, quantum graphity, cellular-automaton and hypergraph rewriting): several *do* join a quantum branch to a curvature/geometry branch conceptually, but we found no case in which that join is delivered as a recompilable, axiom-free proof-kernel artifact rather than as analytic or numerical mathematics.
- **Machine-checked physics/mathematics** (Isabelle/HOL Minkowski and relativistic kinematics; verified PDE and numerical analysis; the four-colour, odd-order, and Kepler formalizations): these deliver verified kernels, but on objects that are either pure-mathematics theorems or the special-relativistic/kinematic layer — not a single equation whose readouts include *dynamical curvature* alongside the quantum branch.
- **Lattice QFT**: delivers exact discrete quantum-plus-special-relativity computation, but on a non-dynamical lattice geometry; the curvature branch is absent by construction.

The intersection of “machine-checked kernel” with “both branches, one equation, dynamical curvature” is where our search returned empty. [Dr]

Retract-on-counterexample commitment. We commit, in print, to retract or narrow C1 upon exhibition of a single machine-checked, axiom-free equation whose readouts demonstrably include both the quantum-dynamics branch and a dynamical-curvature branch. The bar for the counterexample is exactly our own bar (§13): a recompilable kernel artifact, not a paper claim. We ask readers who know of such an artifact to send it; the claim is designed to be killed by one, and we would count that a good outcome for the field. [Dr]

What C1 does and does not assert. C1 does *not* claim our physics is deeper, more predictive, or more complete than any programme in §9 — it is not, and §9 says so. C1 claims only that the specific *join*, at the specific *tier* of a machine-checked axiom-free kernel, is absent elsewhere and present here (N3–N15, §2–§6). It is a claim about a gap in the verified-artifact literature, not about the ceiling of physics.

11 Honest elevation and future work

The present contribution is, on the physics side, largely *re-derivation-as-readout*: the quantum, diffusive, telegraph, horizon, and curvature contents are known physics recovered here as exact machine-checked readouts of one discrete equation, not new predictions — with the single exception of the eigenmode-relaxation-time floor τ_{rel} predicted from the bounded spectrum (§8), which is a genuine quantitative falsifiable statement. We therefore close by stating, plainly and thesis-consistently, the path that would *elevate* the tier of this work — and the directions we refuse as elevation.

The tier-raising path. The method here is already at world rigor: axiom-free over $\mathbb{Q}$, `coqc`-checked, `Print Assumptions Closed` (§13). What is *not* yet at the top tier is the physics, which is re-derivation plus one prediction. The honest ladder up, in order of increasing force, is:

1. **A quantitative falsifiable prediction.** *Done, once*: the τ_{rel} floor from the bounded spectrum (§8) is a sharp, discrete, falsifiable number, not a re-derivation. The elevation target is to add *more* such discrete predictions — not more re-derivations. [Open]
2. **One dimensionless value computed from graph structure.** Compute a single dimensionless quantity (a ratio, an index, a critical exponent) directly from the retained-distinction graph’s structure, machine-checked over $\mathbb{Q}$, and compare it to an independently measured dimensionless number. A dimensionless target keeps us clear of the `OPEN_CONSTANTS` absolute IR scales. [Open]
3. **Independent reproduction.** The artifact is built to be recompiled by others (§13); an independent party recompiling the kernel and confirming the [Th-coqc] claims moves the work from “author-verified” to “community-verified.” This is the cheapest elevation and we actively solicit it. [Open]
4. **A dynamical-GR number match.** The strongest honest elevation would be a *discrete* quantity computed in this kernel that matches a measured general-relativistic number — obtained by sharpening the discrete geometry ($K L_R$, the horizon knife-edge), *not* by taking a continuum limit. [Open]

How we will *not* elevate. Thesis consistency binds the ladder above. We elevate by sharpening *discrete* predictions — never by chasing the refused continuum, and never by inventing a value. Concretely: we will not claim a continuum Einstein-equation limit as an achievement (the continuum is [Refused], §7); the absolute values of M, D, K, τ_c remain [Open] until a discrete derivation supplies them (`OPEN_CONSTANTS`, §12) — no anchor, none claimed; and we will not report a match manufactured by tuning a free parameter to a target.

An elevation that required the continuum, or an unanchored constant asserted as if derived, is not an elevation we will accept — it would trade the one thing this programme has (a verified discrete kernel) for the one thing it refuses.

Tier-honest summary of standing. The *method* is world-rigor (axiom-free $\mathbb{Q}$, `coqc 8.20.1`, `coqchk`); the *physics* is largely re-derivation-as-readout, with exactly one new quantitative falsifiable prediction (the τ_{rel} floor) and no other new physical predictions; and the manuscript is a preprint, *not* yet peer reviewed. We claim the join of C1 (§10) at the tier of a machine-checked kernel, and we claim nothing above that tier. [Dr]

12 The honest remainder: [Open] extensions versus [Refused] non-readouts

[Open] — genuine positive research targets (not gaps in the closed results). None of N1–N15 is open; the discrete tensorial spine and the curvature chain are closed [Th_coqc]. What is open is strictly the following forward extensions:

- The **full metric-derived Riemann array** R^i_{jkl} in $n \geq 3$ satisfying all symmetries, derived from a general (non-diagonal, non-affine) metric g — not just the $\text{diag}(1, w)$ single component R_{1212} of N11. [Open]
- The **exact non-abelian (group) second Bianchi** in transport/holonomy-cube form. The naive covariant form provably *fails* by the exact Leibniz defect $\times(dA, dB)$ (InfoDiscreteLeibnizObstruction, N15, [Th_coqc]), so the correct statement needs the transport-corrected cube; this is open as a positive construction. [Open]
- The **general parametrized $SO(n)$ rotation family** tied to a metric g ($n = 2$ $SO(2)$ and a witnessed rational $SO(3)$ are closed; the general family is open). [Open]
- The **non-relativistic Schrödinger reduction** of the spine, and the numeric values of the absolute constants M, D, K, τ_c — OPEN_CONSTANTS: there is no external anchor for these absolute values, and none is claimed. (The real/complex root *magnitudes* of the telegraph characteristic are *not* listed here as [Open]: their *sign/regime* over $\mathbb{Q}$ is already [Th_coqc] (§5) and their transcendental *values* are [Refused], below — not an unfinished target.) [Open]

[Refused] — non-readouts declined on thesis grounds (not gaps; §7).

- The **continuum** Riemann tensor, smooth $d^2g, d\Gamma$, and any continuum limit of the spine PDE (I2). [Refused]

- The **orthonormal frame** $\sqrt{g}$ (I1) — proven a non-readout for non-perfect-square g . [Refused]
- **Born-rule uniqueness** (I1). [Refused]
- The **real/complex root magnitudes** of the telegraph characteristic — the transcendental phase/rate *values* living in +reals (I1 non-readout). The *sign/regime* of these roots over $\mathbb{Q}$ is decided and machine-checked (§5, [Th_coqc]); the magnitude *values* themselves are declined as a non-readout, not left [Open]. [Th_coqc]→[Refused]
- **Full continuum GR / Einstein dynamical closure as a target:** the GR content enters only through the discrete KL_R geometry readout and the horizon knife-edge, both [Th_coqc]; the continuum closure is not a goal we accept. [Refused]

Discipline restated: do not tag any N1–N15 result [Open] — the discrete tensorial spine and the curvature chain are closed [Th_coqc]. The absolute constant values are [Open] (no anchor, none claimed); the continuum and $\sqrt{g}$ are [Refused].

13 Reproducibility and code availability

The reader is invited to audit rather than believe. Every [Th_coqc] claim above is a theorem of a Coq 8.20.1 kernel over exact rationals, compiled with exit status 0 and *Closed under the global context* at every `Print Assumptions`; the [finite_diagnostic] witnesses are exact rationals reproduced by `Compute/vm_compute`. The artifact inventory is Table 6. The sources live in the public repository <https://github.com/morrocwi/causal-quantum-gravity> under `formal/`, with a `Makefile`; the single command

`make verify`

recompiles every `formal/*.v` file offline and reproduces every machine-checked claim in this note. Each commit is additionally re-verified end-to-end by continuous integration (`make verify` on GitHub Actions); the CI badge, not any table, is the authoritative live count. Each file is standalone, was numerically pre-verified with exact rational arithmetic (typically 200–400 random draws) before authoring, and states its own `Print Assumptions` profile.

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Code Availability. The Coq sources of every machine-checked claim, maintained at <https://github.com/morrocwi/causal-quantum-gravity> (formal/, with a `Makefile` and reproduction commands), constitute the artifact and reproduce every machine-checked claim offline; each file compiles under `coqc` 8.20.1 with exit status 0 and `Print Assumptions = Closed under the global context`. Every commit is additionally re-verified end-to-end by continuous integration (`make verify` on GitHub Actions); the badge is the authoritative live count.

Node	Object / equation	File Tier	Gate
N1 difference	$\text{christoffel_fd } w j = w(Sj) - w(j)$	InfoD[Th.coqc] . riemann.is.second.difference	riemannCurvature ring identity, no $h \rightarrow$ 0
N2 L_R	root graph-Laplacian = second difference of the field	core; [Th.coqc] riemann_fd = $w(SSj) -$ $2w(Sj) +$ $w(j)$	axiom- free; the spatial geome- axis
N3 spine	$M\partial_{tt}\Phi + D\partial_t\Phi + KL_R\Phi = 0$	InfoS[Th.coqc] . thinning	reference read- outs of one equa- tion
N4 telegraph	spine is telegraph, $M = 1/(2\tau_c)$	core [Th.coqc] InfoTelegraph	mass iden- tity ex- act over $\mathbb{Q}$
N5 crossover	$\text{disc} = D^2 - 4MK\lambda; \lambda_c = D^2/(4MK)$	InfoT[Th.coqc] . char.complete.square	complete of- square ex- act; both signs wit- nessed
N6 horizon	telegraph disc = core horizon disc; crossover = agency edge	InfoT[Th.coqc] . disc.is.spine.discr	definition to core InfoF
N7 curvature (scalar)	affine $\Rightarrow$ flat; Christoffel $\neq$ curvature	InfoD[Th.coqc] . affine.is.flat	un- affine wit- ness rieman $2 \neq$ 0
N8 curvature (2-index)	R_{xy} = center of $[X, Y] = a \cdot b$ (Heisenberg, division-free)	InfoD[Th.coqc] . plaquette.curvature.z	adjoint dir flat; kl - antisym ex- act
N9 metric-compatible	$\mathfrak{so}(3)$ generator antisym; $[\text{asym } u, \text{asym } w] = \text{asym}(u \times w)$; Jacobi = 1st Bianchi	InfoM[Th.coqc] . bianchi_jacobi	curvature antisym + al- ge- braic Bianch divisio free
N10 rational isometry	$SO(2)$ Cayley $N^\top N = (1 + t^2)^2 I$; $SO(3)$ holonomy $\neq I$	InfoR[Th.coqc] . InfoRationalS03Curvature	rational or- thog- o- nal trans- port, no $\sqrt{}$
N11 metric-derived	Levi-Civita $\Gamma^2_{12} = \frac{1}{2} dw/w$ rational; R_{1212} rational	InfoM[Th.coqc] . christoffel.G212.rational	needs only g^{-1} (ra- tio-)

Domain	Problem size	Key result	Sparse:dense mem.	Wall-clock	Exact Coq cross-check (tier)
Cosmology	103,823 nodes (47^3 web)	horizon $\lambda_c = 1$; split 40/40 OSC/DECAY	8.6 MB : 80.3 GB = 9606 ×	96.9 s	disc(λ_c) = 0, disc(1, 1, 1, 1) = −3 [Th.coqc]
Quantum	ring $n = 400$ (398 modes)	crossover $\lambda_c = 1$: 265 OSC / 133 DECAY	sparse L_R (nnz 800)	0.4 s	disc(1, 1, 1, 1) = −3, disc(1, 4, 1, 1) = +12 [Th.coqc]
Matter	path $n = 201$; ring $n = 12$	mass = $D/2M$; kink $w = 2.888$ ($\sim \sqrt{2K}$), 100% loc.	dense-free ($n \leq 201$)	0.3 s	mass(1, 1) = 1/2 [Th.coqc]; 7/7 pass
Curvature	nodes $0 \dots 7$, exact $\mathbb{Q}$	$R^{\text{fd}}(n^2) = 2$, $R_{1212} = -\frac{1}{4}$, [X, Y]=15	division-free exact	0.2 s	5 Coq witnesses, 15/15 pass [Th.coqc]

[Th.coqc] = value matches a Coq 8.20.1 theorem bit-for-bit (exact rational); [finite_diagnostic] = finite numeric readout. No dense N^2 array is ever formed; the continuum is never a target.

Table 5: Flagship physics benchmarks (finite / discrete / sparse; the continuum is refused).

File	Tag	Contents (selected)	Checks	SHA-256 prefix (16 hex)
InfoDiscreteRiemannCurvature[Th.coqc].attempt.v	[Th.coqc]	Discrete Riemann curvature as the metric’s finite second difference of the Christoffel; exposes the smuggled continuum gradient.	5	33af18de36085e02
InfoDiscreteGaussBonnet[Th.coqc].attempt.v	[Th.coqc]	Total curvature telescopes to a boundary Christoffel term; closed loop $\Rightarrow$ zero total curvature (finite Gauss–Bonnet).	3	1620b0177ccd0fb0
InfoDiscreteRiemannCommutator[Th.coqc].attempt.v	[Th.coqc]	Genuine 2-index curvature = parallel-transport holonomy / Heisenberg group commutator (matrix non-commutativity).	5	19af721b717a05a8
InfoConnectionFromFrame[Th.coqc].attempt.v	[Th.coqc]	Ties the curvature-commutator to a group-valued frame field; pure-gauge connection is provably flat.	3	4d9986de7bf2b126
InfoMetricCompatibleCurvature[Th.coqc].attempt.v	[Th.coqc]	Metric-compatible curvature: ij -antisymmetry and algebraic (first) Bianchi at the Lie-algebra level, division-free.	4	a4ded27578566594
InfoRationalIsometry[Th.coqc].attempt.v	[Th.coqc]	A rational isometry ($M^\top M = I$) exists over $\mathbb{Q}$ division-free via the Cayley transform ($SO(2)$ gap closed).	4	a4d868858d190100
InfoRationalSO3Curvature[Th.coqc].attempt.v	[Th.coqc]	Capstone: two rational $SO(3)$ rotations whose holonomy is a nonzero rational metric-compatible curvature, division-free.	4	55c5d53ccdaf8c24
InfoMetricDerivedCurvature[Th.coqc].attempt.v	[Th.coqc]	Levi-Civita Christoffel/curvature is a rational readout (g^{-1} only, no frame square root, no 11).	3	0dd29cb5f7f4f1fc
InfoDiscreteSecondBianchi[Th.coqc].attempt.v	[Th.coqc]	Differential (second) Bianchi $dF = 0$ for the abelian case, exact finite-difference commuting identity ($ddA = 0$).	3	6ab41c92d8ef89e9
InfoRiemannPairSymmetry[Th.coqc].attempt.v	[Th.coqc]	Pair symmetry $R_{ijkl} = R_{klij}$ derived (not posited) from the two antisymmetries + first Bianchi.	1	713c46c8d44c361a
InfoMetricFrameNonReadout[Th.coqc].attempt.v	[Th.coqc]	The orthonormal frame $\sqrt{g}$ is an 11 non-readout (irrational) for a rational metric — refused, not fillable.	3	cdade9e6a33910b2
InfoDiscreteLeibnizObstruction[Th.coqc].attempt.v	[Th.coqc]	The discrete Leibniz defect obstructing non-abelian covariant Bianchi is exactly cross(dA, dB).	2	5823bdcef3dc81aa
InfoTelegraphCrossover[Th.coqc].attempt.v	[Th.coqc]	Dynamical quantum $\leftrightarrow$ classical crossover = the telegraph discriminant; its critical point is the spine knife-edge.	4	9339a60c7c319b9e
InfoTelegraphHorizonUniformity[Th.coqc].attempt.v	[Th.coqc]	The telegraph crossover <i>is</i> the black-hole horizon = agency knife-edge: one λ_c on the one spine.	4	8c1f783b858c6880
Totals: 14 files	[Th.coqc]	Discrete-curvature $\rightarrow$ telegraph-horizon Coq stream, all $\mathbb{Q}$ -only axiom-free.	48	14 files

Table 6: Artifact inventory — authoritative copies at `formal/` of <https://github.com/morrocwi/causal-quantum-gravity>; SHA-256 prefixes (first 16 of 64 hex characters) of `sha256sum` on each `.attempt.v` source. Each file is standalone and compiles under `coqc 8.20.1` with exit status 0. “Checks” counts `Print Assumptions` commands; every one reports *Closed under the global context* (axiom-free over $\mathbb{Q}$). Every file was numerically pre-verified with exact rational arithmetic (typically 200–400 random draws) before authoring.